Lab 08 – Passive OT IDS with Zeek

Industrial Security (Bachelor) • Maps to Section 08 • 45–60 min

LAB 08

Goal

Show that a passive ICS-aware IDS detects the same Modbus injection you ran in Lab 04 without ever sending a packet itself. Tune one false positive at the end.

Step 1 – Stack and baseline traffic

The Zeek container shares the OpenPLC network namespace so it sees all Modbus traffic on port 502. From `bachelor/labs/_stack`, generate one minute of baseline polls:

```
cd bachelor/labs/_stack
docker compose exec -d student python3 /labs/scripts/modbus_poll_loop.py
sleep 60
docker compose exec student pkill -f modbus_poll_loop
```

Inspect host-side logs (still from `_stack/`):

```
ls zeek-logs/
tail zeek-logs/modbus.log
```

Observation: you should see one Modbus request line per poll. Note the function code and the source/dest pair.

Step 2 – Trigger the injection

```
docker compose exec student python3 /labs/scripts/modbus_inject.py
tail zeek-logs/notice.log
```

Within a few seconds, a notice should land in `notice.log` tagged `ModbusWatch::Unsolicited_Write`.

Step 3 – Tune one false positive

Edit `bachelor/labs/_stack/zeek/local.zeek` so writes from the student container (`172.28.0.20`) are no longer treated as notices – assume the engineering workstation is sanctioned. Reload Zeek:

```
docker compose restart zeek
```

Re-run the injection: the notice should now be suppressed while normal polls keep being logged.

Deliverable

Hand in: `notice.log` excerpts before and after tuning, your modified `local.zeek`, and a 5-line answer to: "*Who would receive this alert in a real SOC, and what is their first action?*"

Caution

Lab containers only. Zeek here passively observes a private Docker network; running it on a production network requires explicit authorisation and usually a SPAN/TAP feed, not a shared netns.